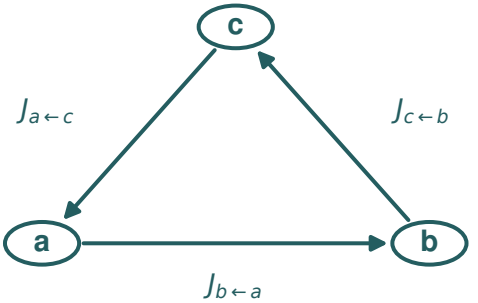
Exact coordinate transport and chosen connection data

The same directed loop carries two different kinds of transport.

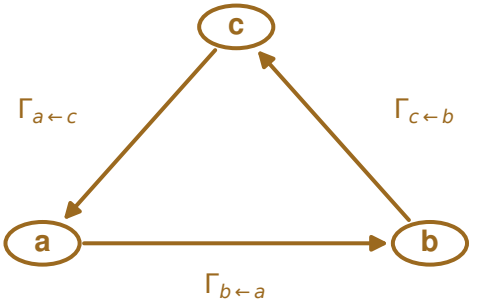
Canonical coordinate transport



$$J_{a\leftarrow c}J_{c\leftarrow b}J_{b\leftarrow a} = \mathrm{id}$$

Exact analysis and synthesis force the factors to telescope.

Chosen connection



$$H_{c,b,a} = \Gamma_{a\leftarrow c}\Gamma_{c\leftarrow b}\Gamma_{b\leftarrow a}$$

$$H^{G} = G_a H G_a^{-1}$$

Canonical loops are identities. Connection holonomy is additional data and changes by conjugation.